

Robot Motion Planning and Implementation.

By

Hove, Mamana, Yann, Reuben, Marvellous, Chilufya

AIMS

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Introduction

Plan

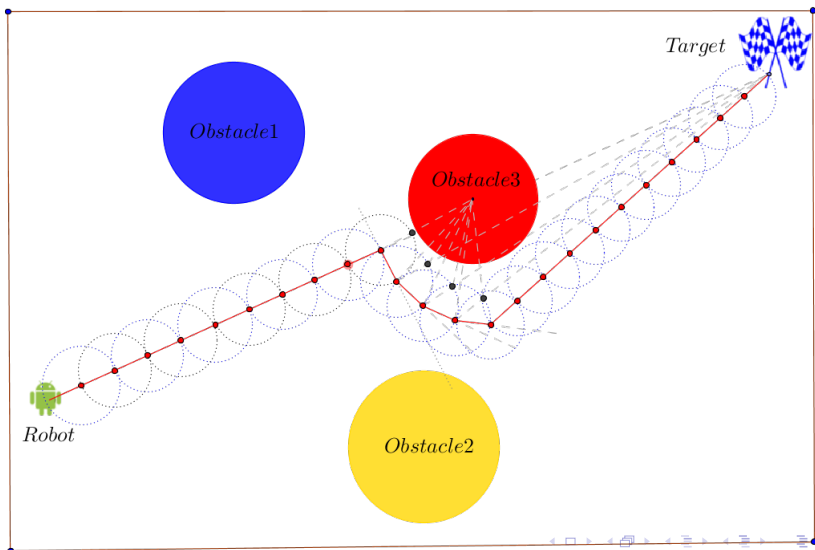
- Create a robot that moves towards a target in any given environment, while avoiding various obstacles .
- To do this, two algorithms were used.
 - 1 Intuitive idea related to the tangent bug algorithm.
 - 2 Rapidly-exploring Random Tree (RRT) based algorithm.

The Intuitive Idea Algorithm

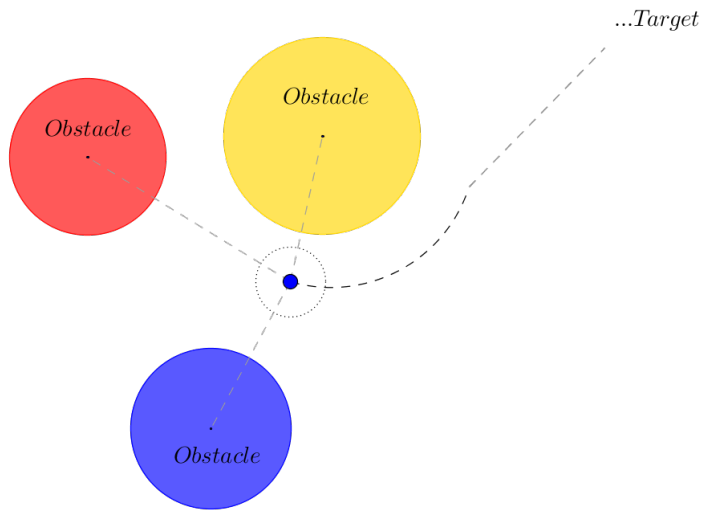
Initial set-up

- Create the window with the fixed and random obstacles, robot and target at targeted positions using pygame library.
- The object is bounded by set of points (r). Note the radius (R) of each obstacle.

The Intuitive Idea Algorithm



The Intuitive Idea Algorithm

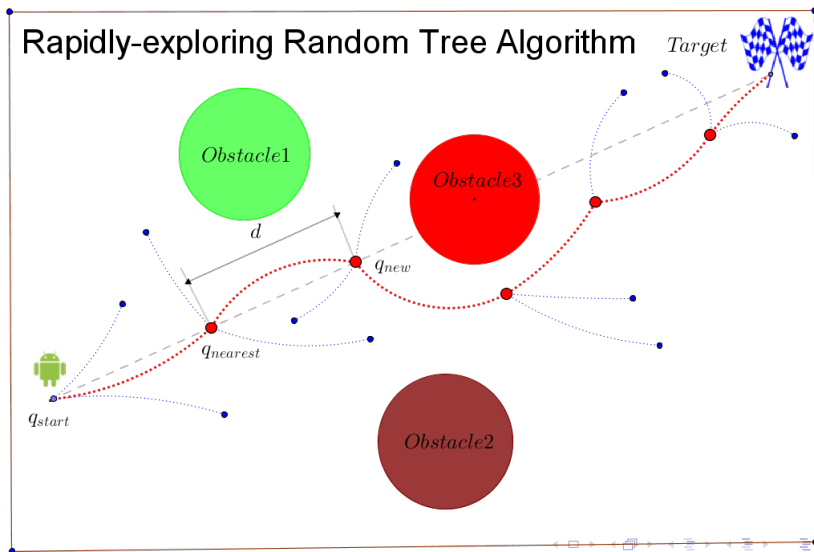


The Intuitive Idea Algorithm

Describing the motion of the robot

- The robot is set into motion.
- Let Ω = set of obstacles, $m = \min(\text{Object}, X)$, i.e. the distance between the object and the obstacle X for $X \in \Omega$.
- If $m \geq R + 2r$, the robot (object) move forward along the direction from the current position of the robot to the target with distance ϵ .
- If not, then the robot moves along the direction which is normal to the direction from the current position of the robot to the target.
- This process is repeated until the robot reaches its target.

Rapidly-exploring random tree algorithm



Rapidly-exploring random tree algorithm

This algorithm creates a tree in a way that provides good coverage for robot's motion, it's only after the path being found that the robot will begin moving following the path generated by the algorithm. The algorithm is given below.

- Generate random point within window size.
- Find the nearest neighbour to that point.
- Select an action from the neighbour that heads towards the random point.
- Create a new point based on the outcome of the action applied to the neighbour.
- Add the new point to the tree, and connect it to the neighbour.
- The above steps are then repeated in a loop until the target point is reached.

Limitations and Work remaining

- The algorithm works only for obstacles of same geometrical properties, so our next goal will be to generalise the geometrical form of the obstacles;
- For the first algorithm, make the robot turn towards the leftside away from the obstacle;
- For the RRT based algorithm, we would like to customise the algorithm to get, if possible, an hybride algorithm.

THANK YOU